

Phosphate plant -- Sensors

Section report: Sensors

Site	Atlas Phosphate Complex (:atlas_phosphate)
Reporting period	2026-01-01T00:00:00 to 2026-12-31T23:00:00 (365 days, 8568 operating hours)
Product	Dap from 29.0 % P2O5 ore, 1713.0 kt of product
Framework scope	ISO 50001:2018 energy management; ISO 14001:2015 environmental management; ISO 46001:2019 water efficiency; ISO 9001:2015 quality management; GHG Protocol corporate standard; EU Industrial Emissions Directive 2010/75; US EPA 40 CFR 63 subpart AA; IFA/EFMA BAT for phosphate fertilisers; Phosphogypsum stack management code
Provenance	Design revision A seed 20260101 8760 hourly intervals of 69 signals generated 2026-09-23T15:43:11.110 basis :screening_plant_model

The inferential sensors: process, vision and acoustics

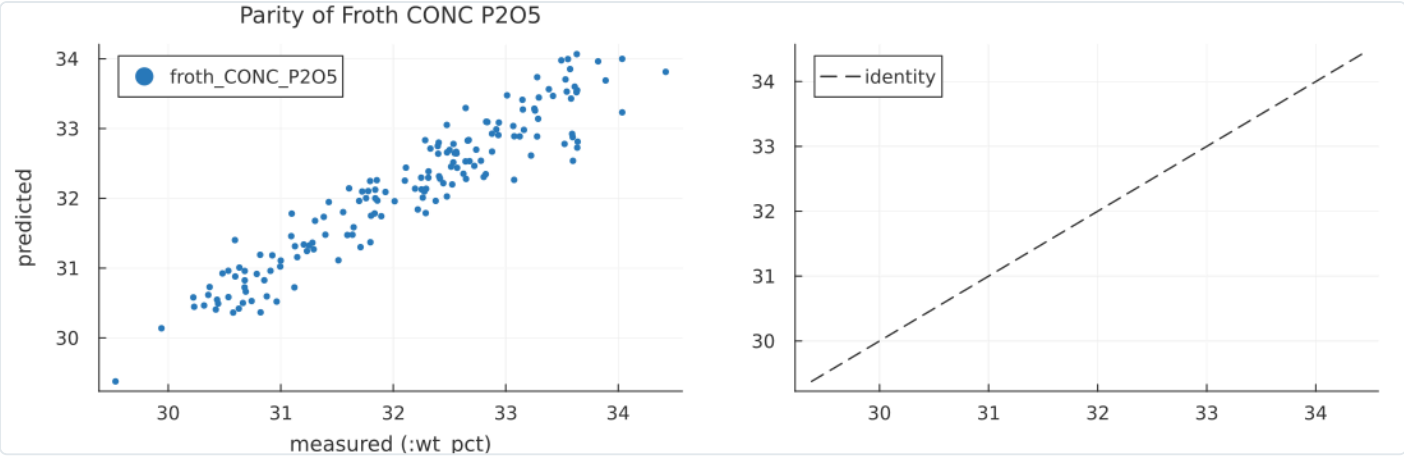
THE REGISTER OF THE INFERENTIAL SENSORS

Quantity	Value									
Sensors	14 inferential models, of which 4 read an image or a sound									
Median R2	0.57364									
Best sensor	:pump_PUMP_VIBRATION									
Status	:at_risk									

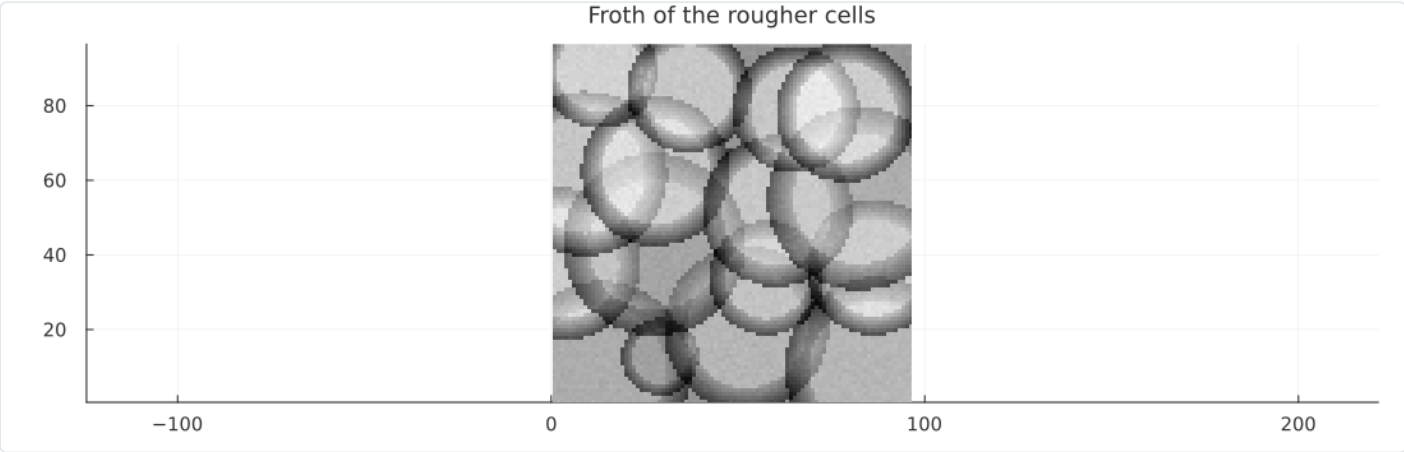
Sensor	Infers	Modality	Unit	R2	RMSE	Bias	In domain %	Drift	Status	Inputs
:pump_PUMP_VIBRATION	:PUMP_VIBRATION	:acoustic	:count	0.9896	0.073906	-0.0097	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:mill_MILL_POWER	:MILL_POWER	:acoustic	:kw	0.9423	83.7837	-23.52	n/a	0	Healthy	rms, crest, kurtosis, band_low, band_mid, band_high, band_veryhigh, centroid, dominant_hz, cavitation_index, tonal_ratio
:process_MILL_POWER	:MILL_POWER	:process	:kw	0.93678	106.9652	3.45	100	1.38	Healthy	ROM_FEED, SLURRY_DENSITY, MILL_FEED_WATER
:process_STEAM_FLOW	:STEAM_FLOW	:process	:t_ph	0.93115	1.6977	-0.0218	100	-0.0087	Healthy	EVAP_FEED_FLOW, STRONG_ACID_P2O5, EVAP_VACUUM
:process_CONC_P2O5	:CONC_P2O5	:process	:wt_pct	0.91007	0.30973	0.0084	100	0.0034	Healthy	FLOT_FEED_P2O5, CYCLONE_P80, REAGENT_FLOW
:froth_CONC_P2O5	:CONC_P2O5	:visual	:wt_pct	0.89295	0.356853	-0.0189	n/a	0	Healthy	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:process	:wt_pct	0.69479	0.175129	0.0023	100	0.0009	Healthy	SLURRY_SG, ATTACK_TEMP, FREE_SO4, WASH_WATER_FLOW
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:process	:wt_pct	0.45249	0.19828	0.0007	100	0.0003	Healthy	EVAP_TEMP, STEAM_FLOW, EVAP_VACUUM
:rock_CYCLONE_P80	:CYCLONE_P80	:visual	:um	0.39939	3.044	0.5433	n/a	0	Degraded	img_mean, img_std, img_entropy, img_band_low, img_band_mid, img_band_high, img_edge_density, img_local_contrast, img_bubble_count, img_bubble_px, img_dominant_f
:process_PRODUCT_N	:PRODUCT_N	:process	:wt_pct	0.33168	0.2647	0.0021	100	0.0009	Healthy	AMMONIA_FLOW, PRODUCT_FLOW, MELT_DENSITY
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:process	:wt_pct	0.21885	0.018076	0.0001	100	0	Healthy	FILTER_RATE, FILTER_VACUUM, WASH_WATER_FLOW, WEAK_ACID_P2O5
:process_FILTER_RATE	:FILTER_RATE	:process	:t_per_m2_d	0.21477	1.165	0.0177	100	0.0071	Healthy	FILTER_VACUUM, WEAK_ACID_P2O5, SLURRY_SG, GYPSUM_FREE_P2O5
:process_CYCLONE_P80	:CYCLONE_P80	:process	:um	0.03116	6.1781	-0.1283	100	-0.0513	Healthy	MILL_POWER, SLURRY_DENSITY, CYCLONE_PRESSURE, MILL_FEED_WATER
:process_ATTACK_TEMP	:ATTACK_TEMP	:process	:deg_c	-4.0e-5	16.3148	0.2679	100	0.1072	Healthy	H2SO4_FLOW, ROCK_FEED, RECYCLE_ACID_FLOW, SLURRY_SG

WHAT THE SENSORS SAY AGAINST WHAT THE PLANT MEASURED

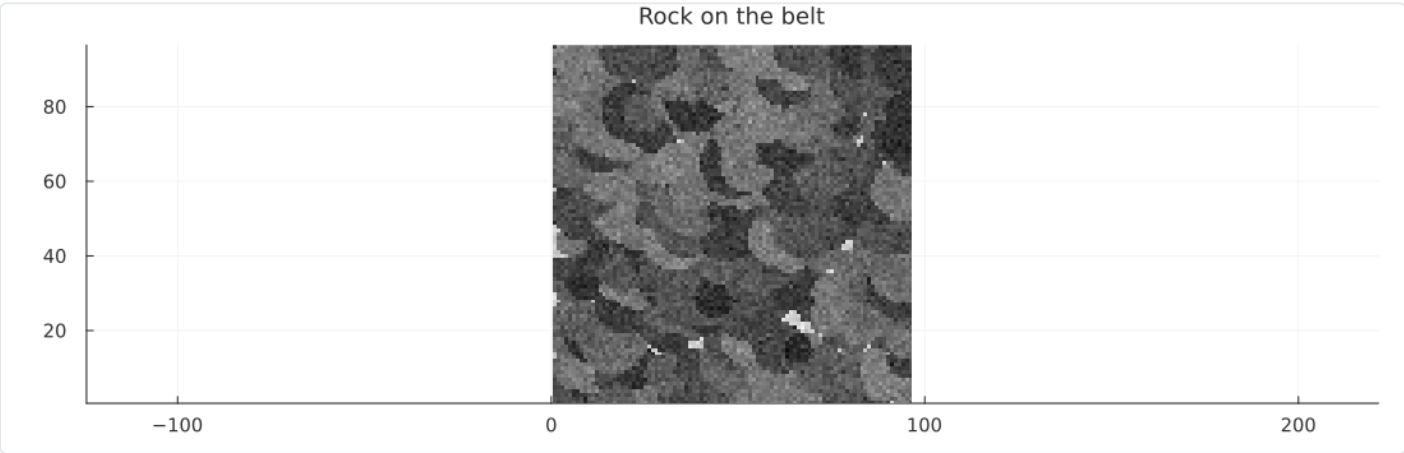
Sensor	Infers	Unit	Estimate	Measured	Difference %	RMSE	Status
:process_STRONG_ACID_P2O5	:STRONG_ACID_P2O5	:wt_pct	55.77	55.85	-0.1382	0.1983	Compliant
:process_WEAK_ACID_P2O5	:WEAK_ACID_P2O5	:wt_pct	28.75	28.7	0.1628	0.1751	Compliant
:process_GYPSUM_FREE_P2O5	:GYPSUM_FREE_P2O5	:wt_pct	0.5028	0.5069	-0.8103	0.0181	Compliant
:process_CYCLONE_P80	:CYCLONE_P80	:um	150.82	149.52	0.8702	6.18	Compliant
:process_MILL_POWER	:MILL_POWER	:kw	5,523.06	5,524.95	-0.0341	106.97	Compliant
:process_PRODUCT_N	:PRODUCT_N	:wt_pct	18.11	18.11	-0.0093	0.2647	Compliant
:process_FILTER_RATE	:FILTER_RATE	:t_per_m2_d	8.76	8.89	-1.46	1.16	Compliant
:process_CONC_P2O5	:CONC_P2O5	:wt_pct	32.4	32.47	-0.2148	0.3097	Compliant
:process_ATTACK_TEMP	:ATTACK_TEMP	:deg_c	77.55	78.18	-0.8028	16.31	Compliant
:process_STEAM_FLOW	:STEAM_FLOW	:t_ph	51.47	51.69	-0.4197	1.7	Compliant



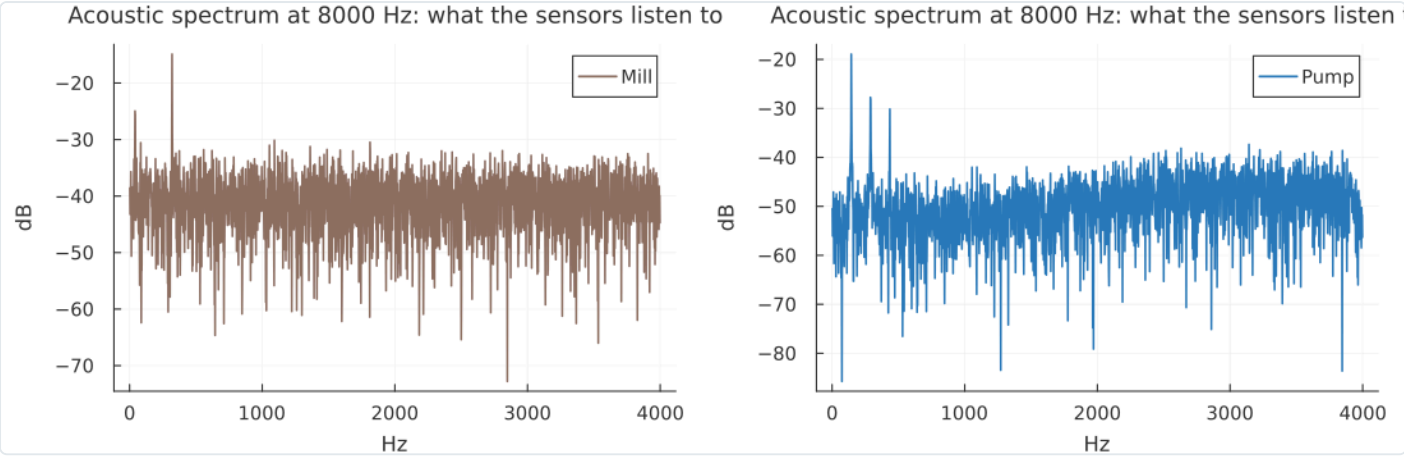
Parity of the froth sensor against the analyser



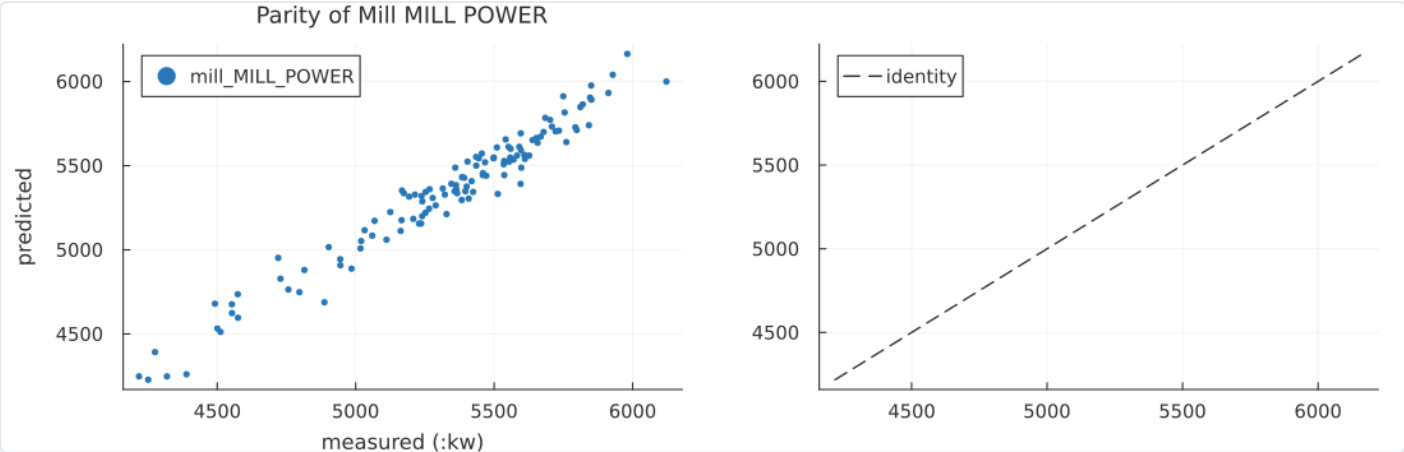
The froth image the visual sensor reads the grade from



The image of the rock the visual sensor reads the grind from



Spectra of the mill and of the pump: the two acoustic channels of the plant



Parity of the acoustic sensor of the mill against the power it draws